

gamma chi estimate.py

```
1 import numpy as np
2 import matplotlib.pyplot as plt
3 from scipy.stats import linregress
4
5 # Step 1: Input data
6 T = np.array([451, 452, 455, 458, 460]) # Temperature in Kelvin
7 chi = np.array([110, 51, 12, 10, 1.5]) # Magnetic susceptibility
8 Tc = 450 # Critical temperature in Kelvin
9
10 # Step 2: Compute reduced temperature  $t = (T - T_c) / T_c$ 
11 t = (T - Tc) / Tc
12
13 # Step 3: Take natural logarithms to linearize the equation
14 ln_t = np.log(t)
15 ln_chi = np.log(chi)
16
17 # Step 4: Perform linear regression:  $\ln(\chi) = \ln(\chi_0) - \gamma \ln(t)$ 
18 slope, intercept, r_value, p_value, std_err = linregress(ln_t, ln_chi)
19
20 gamma = -slope # gamma is -slope of the linear fit
21 chi0 = np.exp(intercept) # chi0 = exp(intercept)
22
23 # Step 5: Print results
24 print(f"Estimated gamma ( $\gamma$ ): {gamma:.4f}")
25 print(f"Estimated chi0 ( $\chi_0$ ): {chi0:.4f}")
26
27 # Step 6: Plotting  $\ln(\chi)$  vs  $\ln(t)$ 
28 plt.figure(figsize=(8, 5))
29 plt.plot(ln_t, ln_chi, 'o', label='Data points')
30 plt.plot(ln_t, intercept + slope * ln_t, 'r-', label='Linear fit')
31 plt.xlabel('$\ln(t)$ ----->')
32 plt.ylabel('$\ln(\chi)$ ----->')
33 plt.title('Linear fit to estimate  $\chi_0$  and  $\gamma$ ')
34 plt.grid(True)
35 plt.legend()
36 plt.show()
37
38 '''
39 Output:
40 Estimated gamma ( $\gamma$ ): 1.6056
41 Estimated chi0 ( $\chi_0$ ): 0.0075
42 '''
```